ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

(i) Pixel pitch p = 5 microm = 0.005 mm. Hence:

$$f_x = f/p = 16/0.005 = 3200 pixels$$

(ii) Sensor width = 1280(0.005) = 6.40 mm. At standoff Z = 400 mm, belt width W = Z(6.40)/16 = 160 mm.

$$Scale = 160/1280 = 0.125 mm/pixel$$

Working: $5\,\mu$m $= 5 \times 10^{-3}$ mm $= 0.005$ mm. So $16/0.005 = 16 \times 200 = 3200$ pixels. For the sensor, $1280 \times 0.005 = 6.4$ mm. Then $W = (400/16) \times 6.4 = 25 \times 6.4 = 160$ mm, and $160/1280 = 0.125$ mm/pixel.

Q1(b) [3 marks]

ADC step = 5/2^12 = 0.0012207 V = 1.2207 mV. Sensor sensitivity is 10 mV/N, so ΔF = 1.2207/10 = 0.12207 N. The smallest resolvable force change is approximately 0.122 N.

Calculation: $2^{12} = 4096$ levels. One step is $5/4096 = 0.0012207$ V, i.e. 1.2207 mV. Dividing by the sensitivity, $1.2207\,\text{mV}/(10\,\text{mV/N}) = 0.12207$ N. The mV units cancel, leaving N.

Q1(c) [3 marks]

For zero-order hold, F = e^(AT) and G = integral_0^T e^(Aτ)B dτ. Here:

$$A = -2, B = 1, T = 0.1s$$

Thus:

$$F = e^(- 0.2) = 0.818731$$

$$G = (1 - e^(- 0.2))/2 = 0.0906346$$

Therefore:

$$x_{k+1} = 0.818731 x_k + 0.0906346 u_k$$

With $AT = (-2)(0.1) = -0.2$, $F = e^{-0.2} = 0.8187308$. For G, $(1 - F)/2 = (1 - 0.8187308)/2 = 0.1812692/2 = 0.0906346$. Thus the input contribution is positive.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differences are poor mainly because the encoder measures angle in degrees instead of radians, and because the velocity signal is always delayed by one full second.

Q3(b) [2 marks]

The Tustin coefficients are y[k]=0.881765y[k-1]+0.0591175(x[k]+x[k-1]). The DC gain is 0.0591175/(1-0.881765)=0.5.

Q3(c) [4 marks]

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. The desired polynomial is $(s+10)^2 = s^2+20s+100$.

Q3(d) [2 marks]

The observer is simpler because it only smooths the measured velocity. The filter is safer when the model is poor.

Question 4 [10 marks]

Q4(a) [2 marks]

Closed-loop characteristic polynomial is $s^2+(a+bk_2)s+(c+bk_1)=s^2+4s+8$, giving poles -2±2j. At DC, $\theta/r = bN_{ff}/(c+bk_1)$, so unity gain requires $N_{ff}=(c+bk_1)/b=(6+2)/1=8$.

Comparing coefficients with $s^2 + 4s + 8$ gives $a + bk_2 = 4$ and $c + bk_1 = 8$. The feedforward gain is chosen from the nominal plant so the reference scaling is unity; here this gives $N_{ff} = 8$. The closed-loop roots of $s^2 + 4s + 8$ are $-2 \pm 2j$.

Q4(b) [2 marks]

$$\theta_{ss} = 0.75 \times 8/(6 + 2) = 0.75,$$

the percentage tracking error is 25%

Q4(c) [4 marks]

With $\dot{\xi}$=r-θ and u=-$k_1\theta$-$k_2\dot{\theta}$+K_Iξ, the homogeneous closed-loop characteristic polynomial is $s^3+(a+bk_2)s^2+(c+$
Desired polynomial: $(s+4)[(s+2)^2+4]=(s+4)(s^2+4s+8)=s^3+8s^2+24s+32$. For a=2,b=1,c=6, matching coefficients gives k_2=6, k_1=18 and K_I=32.

First multiply $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$. Comparing this with the augmented characteristic polynomial gives $k_2 = 6$, $k_1 = 18$, and $K_I = 32$. I checked the coefficients in descending powers of s.

Q4(d) [2 marks]